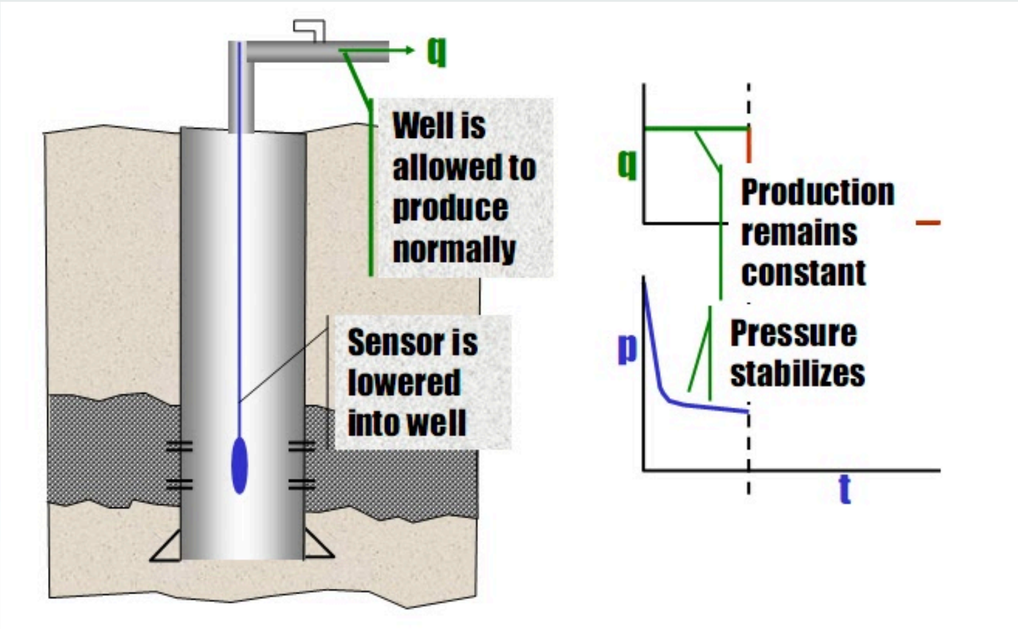

Project
Well Test Analysis of Reservoir
Drawdown, Buildup, Flow-After-Flow & Isochronal Tests
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Introduction

🎯 Objective of the Project

- Perform well test analysis using drawdown, buildup, flow-after-flow, and isochronal tests.
- Estimate reservoir & well parameters: permeability, skin factor, reservoir pressure, and deliverability.



📊 Importance of Well Testing

- Helps us know the reservoir better – its pressure, flow capacity, and condition.
- Tells us how productive a well is and how much it can deliver.
- Shows if the well is damaged near the borehole and if it needs cleaning or stimulation.
- Gives real performance data of the reservoir, not just estimates from logs.
- Helps in planning production – how to lift fluids and optimize output.

Read More . . .

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What Is Well Test

A well test is the process of producing a well under controlled flow and shut-in conditions, and recording the pressure and flow rate response of the reservoir

The analysis of this data provides quantitative information on reservoir properties (permeability, skin factor, reservoir pressure) and well performance (productivity index, deliverability).

General Methodology of Well Testing

- **Create a change in production rate**
(by starting, stopping, or varying the flow rate)
- **Measure the reservoir response**
(response may be pressure, rate, or both)
- **Analyze the reservoir response**
(using theoretical models and equations)



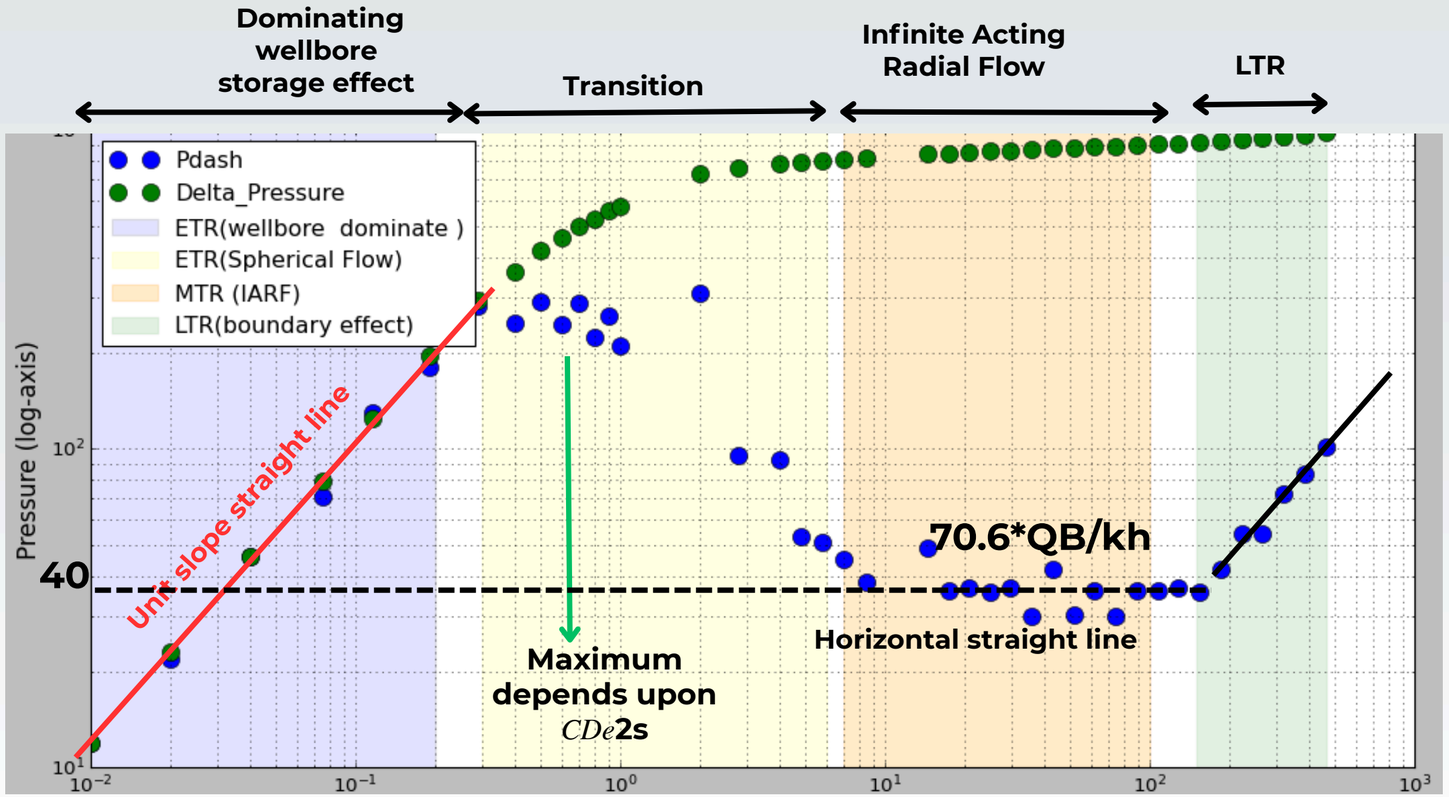
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Drawdown Test

A drawdown test is performed by producing a well at a constant rate and recording the pressure decline over time.

Used to Determine:

- Estimate permeability of the reservoir.
- Calculate skin factor (formation damage or stimulation).
- Identify wellbore storage effects



Log–Log Diagnostic Plot

Interpretation :

- **Unit slope line at early time → wellbore storage dominance.**
- **Horizontal line at derivative → infinite acting radial flow (IARF) region → permeability.**
- **Deviation upwards → sealing boundary (closed reservoir).**
- **Deviation downwards → constant pressure boundary (aquifer support).**
- **Humps/valleys → heterogeneity, fracture, dual-porosity**

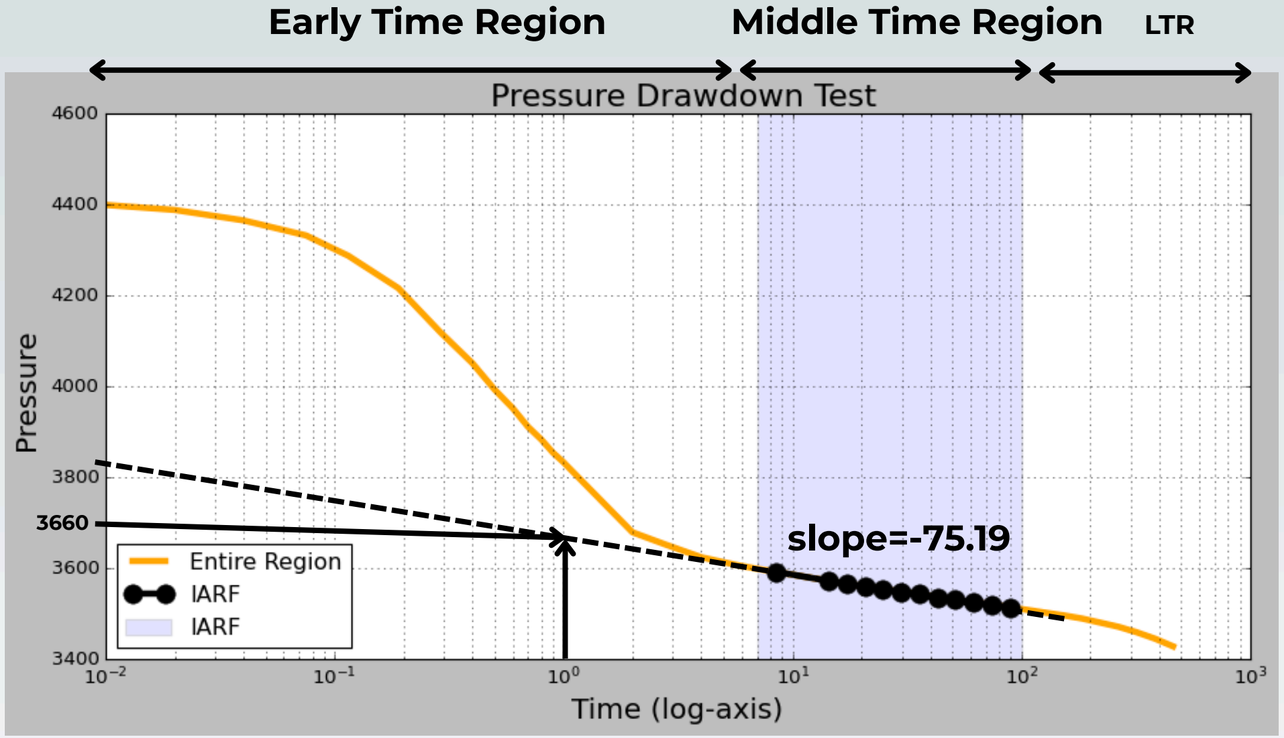
Identifies flow regimes & boundary effects

40=70.6*Q*B/(kh)
k=70.6*250*1.136/(40*69)
k=7.26 md

The permeability of reservoir is 7.26 md

Drawdown Test

Semi-Log Plot



$$Pwf = Pi - \frac{162.6qu}{kh}\left(log\frac{kt}{phie * u * Ct * rw^2} - 3.23 + .87S\right)$$

For permeability

k=(162.6*250*1.136*.8)/(75.19*69)
k=7.12 md

For skin

$$s = 1.151\left(\frac{Pi - Pwf\ at\ 1hr}{m} + 3.23 - log\frac{k}{phie * u * Ct * rw^2}\right)$$

$$s = 1.151\left(\frac{4412 - 3360}{75.9} + 3.23 + log\frac{7.12}{.039 * 0.8 * 17 * 10^{-6} *. 198^2}\right)$$

Interpretation:

- **Early-time deviation → Wellbore storage effect.**
- **Straight line in radial flow region → permeability.**
- **Intercept at Δt = 1 → used to calculate skin factor (s)**
- **Late-time deviation → Boundary effects (sealing fault, aquifer support).**

s=5.40

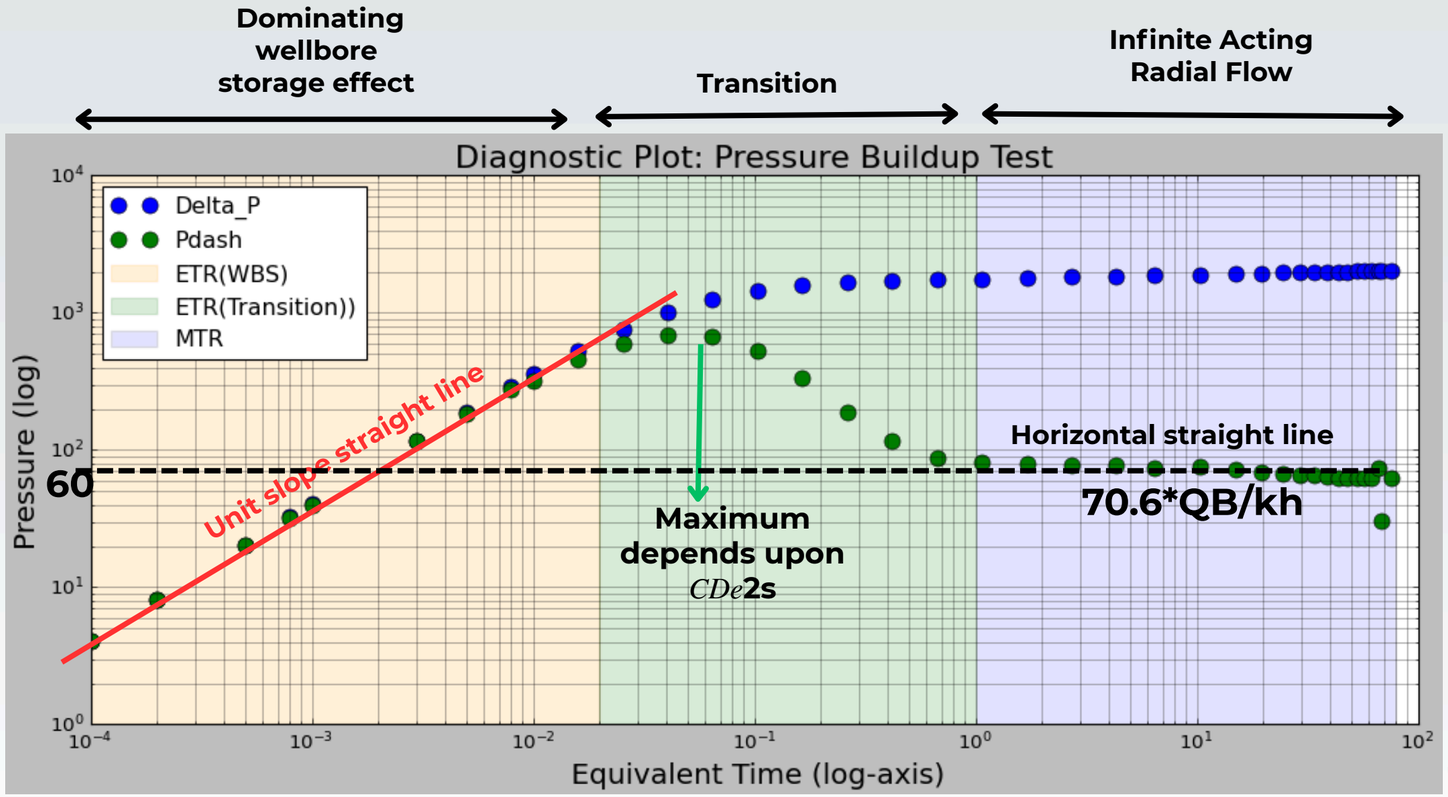
Pressure loss due to skin =0.87*75.19*5.40 =353.25psi

Pressure Buildup Test

A pressure buildup test is performed by shutting in a producing well and recording the pressure increase over time.

Used to Determine:

- Initial Reservoir Pressure or Average Drainage Area Pressure
- Permeability in the well's drainage area
- Skin Factor in the near-wellbore region



Log–Log Diagnostic Plot

Interpretation :

- **Unit slope line at early time → wellbore storage dominance.**
- **Horizontal line at derivative → infinite acting radial flow (IARF) region → permeability.**
- **Deviation upwards → sealing boundary (closed reservoir).**
- **Deviation downwards → constant pressure boundary (aquifer support).**
- **Humps/valleys → heterogeneity, fracture, dual-porosity**

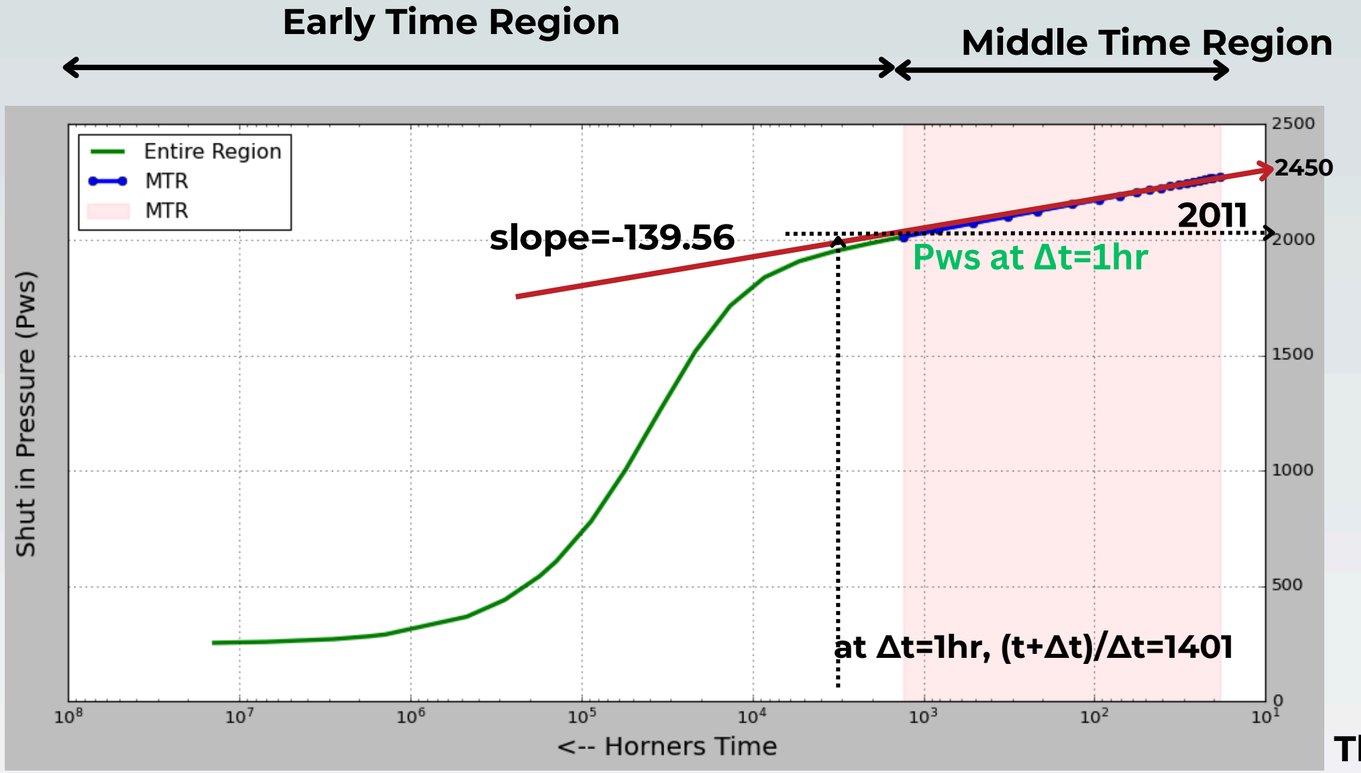
Identifies flow regimes & boundary effects

60=70.6*Q*B/(kh)
k=70.6*600*1.1/(78*60)
k=9.96 md

The permeability of reservoir is 9.96 md

Pressure Buildup Test

Horner Plot :



$$Pwf = Pi - \frac{162.6qu}{kh}\left(\log\frac{t+\Delta t}{t}\right)$$

slope =162.6*qu/kh

k=9.86 md

$$S = 1.151\left(\frac{Pws\ at\ 1hr - pwf\ at\Delta\ t=0}{m}\right| - 3.23 + \log\left(\frac{k}{phie*u*Ct*rw^2}\right)\Big)$$

s=9.77

pressure loss due to skin =.87m*s

=1186.22psi

Interpretation:

- **Straight line in radial flow region → permeability.**
- **Extrapolated intercept → average reservoir pressure.**
- **Early-time deviation → wellbore storage.**
- **Upward/downward curvature → skin effect.**

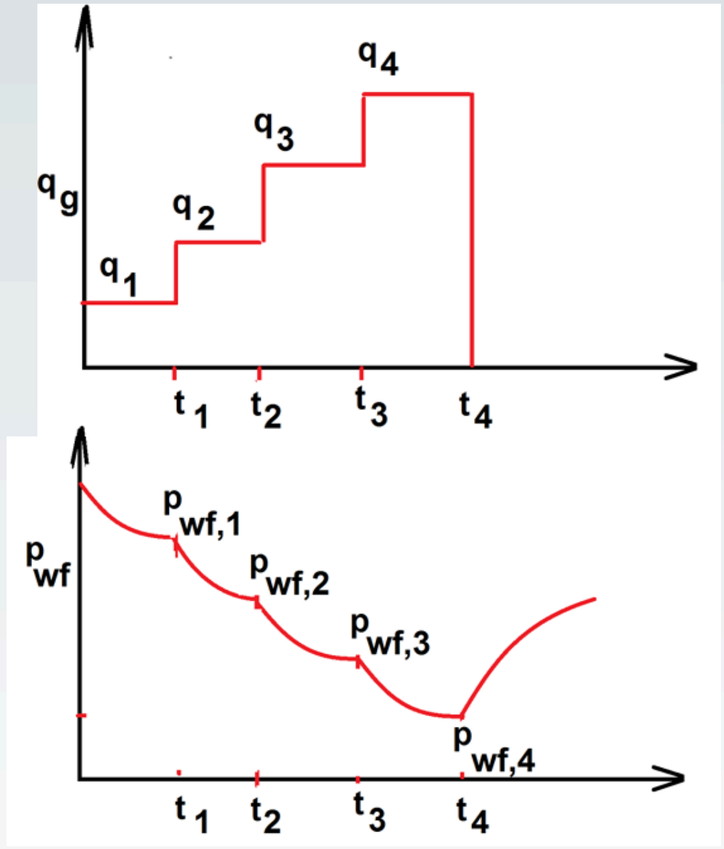
The formation permeability is of only 9.862 md which indicates that the well has poor permeability. The skin factor obtained is positive that means the Zone nearest to the wellbore is likely to be damaged. Since, the skin is positive, there is an additional pressure drop of 1186.222 psia. The well is damaged and a good candidate for stimulation.

Flow After Flow Test

A flow-after-flow test (also called a gas well deliverability test) involves producing a gas well at several stabilized flow rates and measuring the corresponding bottom-hole pressures (pwf). Purpose → To establish a relationship between flow rate (q) and pressure drawdown (pi – pwf).

Methodology

1. **Produce the well at a constant flow rate until stabilized pressure is reached.**
2. **Record bottom-hole pressure.**
3. **Repeat at increasing flow rates.**
4. **Plot data (q vs Δp^2) → obtain Inflow Performance Relationship (IPR).**



Interpretation

- **Empirical Approach**

$$Q_g = C\left(\bar{p}^2 - p_{wf}^2\right)^n$$

- **Theoretical Approach**

$$\bar{p_r}^2 - p_{wf}^2 = aQ_g + bQ_g^2$$

Flow After Flow Test

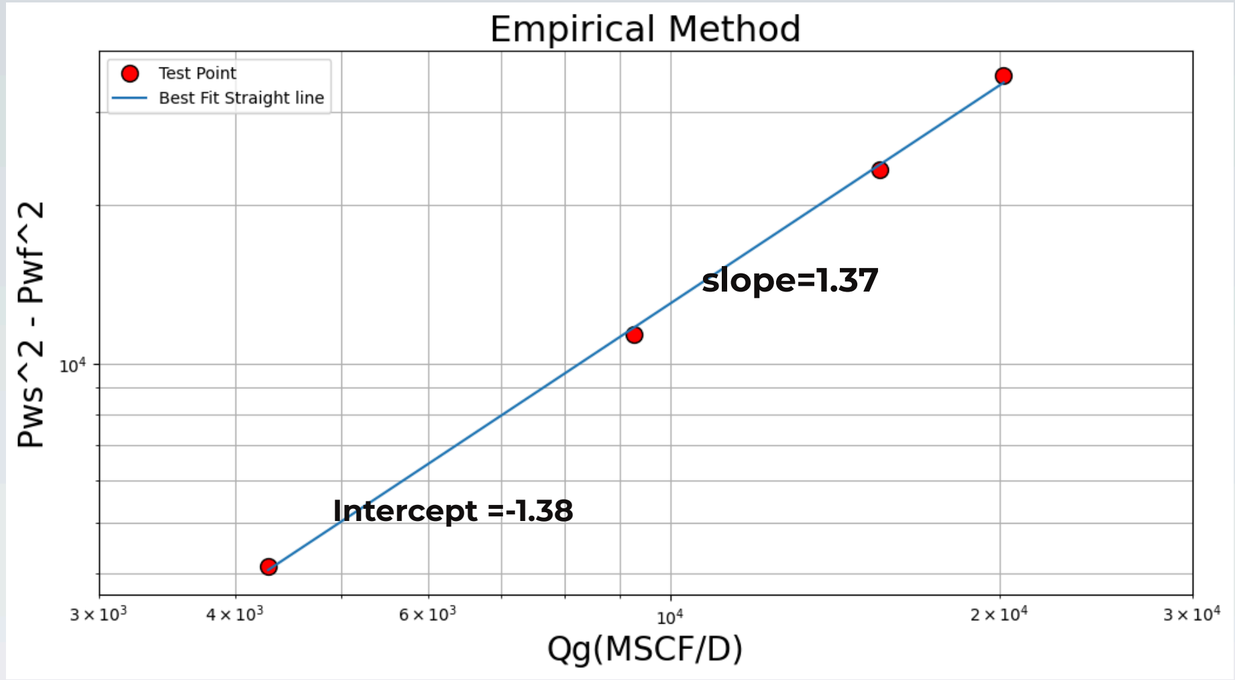
- **Empirical Approach**

$$Q_g = C\left(\bar{p}^2 - p_{wf}^2\right)^n$$

$$\log(Q_g) = n\log\left(\bar{p}^2 - p_{wf}^2\right) + \log(C)$$

$$\log\left(\bar{p}^2 - p_{wf}^2\right) = \frac{1}{n}\log(Q_g) - \frac{\log(C)}{n}$$

$$C = \frac{Q_g}{\left(\bar{p}^2 - p_{wf}^2\right)^n}$$



n=1/slope
n=0.728
c=10^(-intercept*n)
c=10.06

Deliverability equation

$$Q_g = 10.6\left(\bar{p}^2 - p_{wf}^2\right)^{.728}$$

Calculating Absolute open Flow by putting Pwf=14.7 psia

AOF= 63370.703 MSCFD

Flow After Flow Test

- **Theoretical Approach**

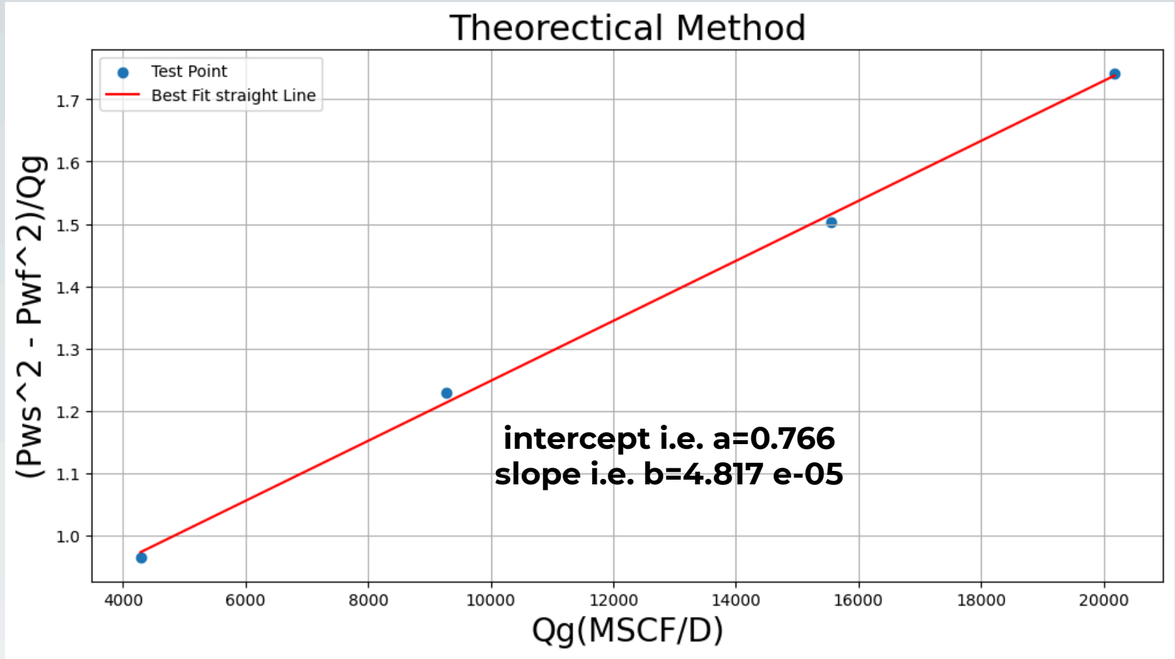
$$\bar{p}_r^{\,2} - p_{wf}^2 = aQ_g + bQ_g^2$$

$$\left(\frac{\bar{p}_r^{\,2} - p_{wf}^2}{Q_g}\right) = a + bQ_g$$

a=0.766
b=4.817 e-05

Deliverability equation

$$\bar{p}_r^{\,2} - p_{wf}^2 = \mathbf{0.766}\ Q_g + \mathbf{4}X10^{-5}Q_g^2$$



Calculating Absolute open Flow by putting Pwf=14.7 psia
AOF= 51359.935 MSCFD

Why the Difference?

Absolute Open Flow (Theoretical Method): 51359.935 MSCFD

Absolute Open Flow (Empirical method) is 63370.703 MSCFD

Difference arises due to:

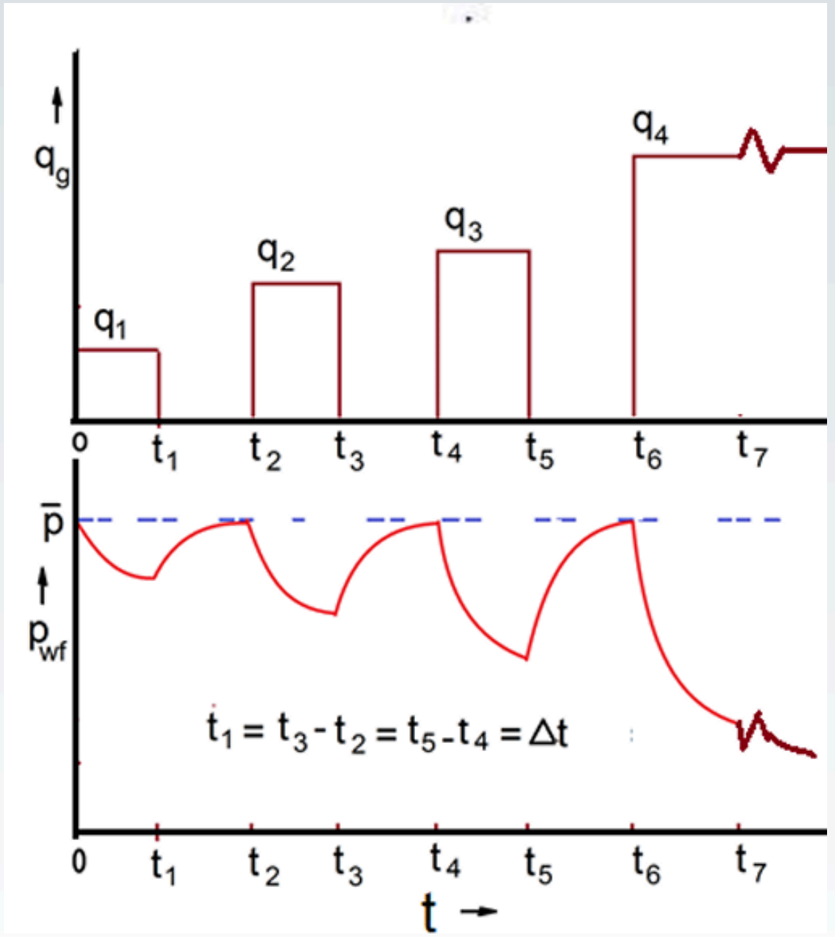
- Skin factor / formation damage.
- Non-Darcy (turbulent) flow.
- Multiphase effects (water or condensate).
- Test duration not long enough to reach stabilization.
- Assumptions of homogeneity in theoretical models.

Isochronal Test

To determine the deliverability of the wellbore without waiting long for the flow to stabilized at a given flow rate as we saw in flow after flow test.

Note: This test is suitable for reservoir with low permeability where it is practically not possible to achieve stabilized conditions

1. **Flow Periods:**
   - The well is flowed at several different constant rates (e.g., q_1, q_2, q_3).
   - Each flow period has the same fixed duration (the isochronal period, usually 1–4 hours).
   - Flow is stopped before full stabilization.
2. **Shut-in Periods:**
   - After each flow period, the well is shut-in long enough to reach initial reservoir pressure again.
3. **Final Stabilized Flow:**
   - At the end, a long flow period is carried out to approximate stabilized conditions.



Interpretation

- **Empirical Approach**

$$Q_g = C\left(\bar{p}^2 - p_{wf}^2\right)^n$$

- **Theoretical Approach**

$$\bar{p}_r^{\,2} - p_{wf}^2 = aQ_g + bQ_g^2$$

Flow After Flow Test

- **Empirical Approach**

$$Q_g = C(\bar{p}^2 - p_{wf}^2)^n$$

$$\log(Q_g) = n\log(\bar{p}^2 - p_{wf}^2) + \log(C)$$

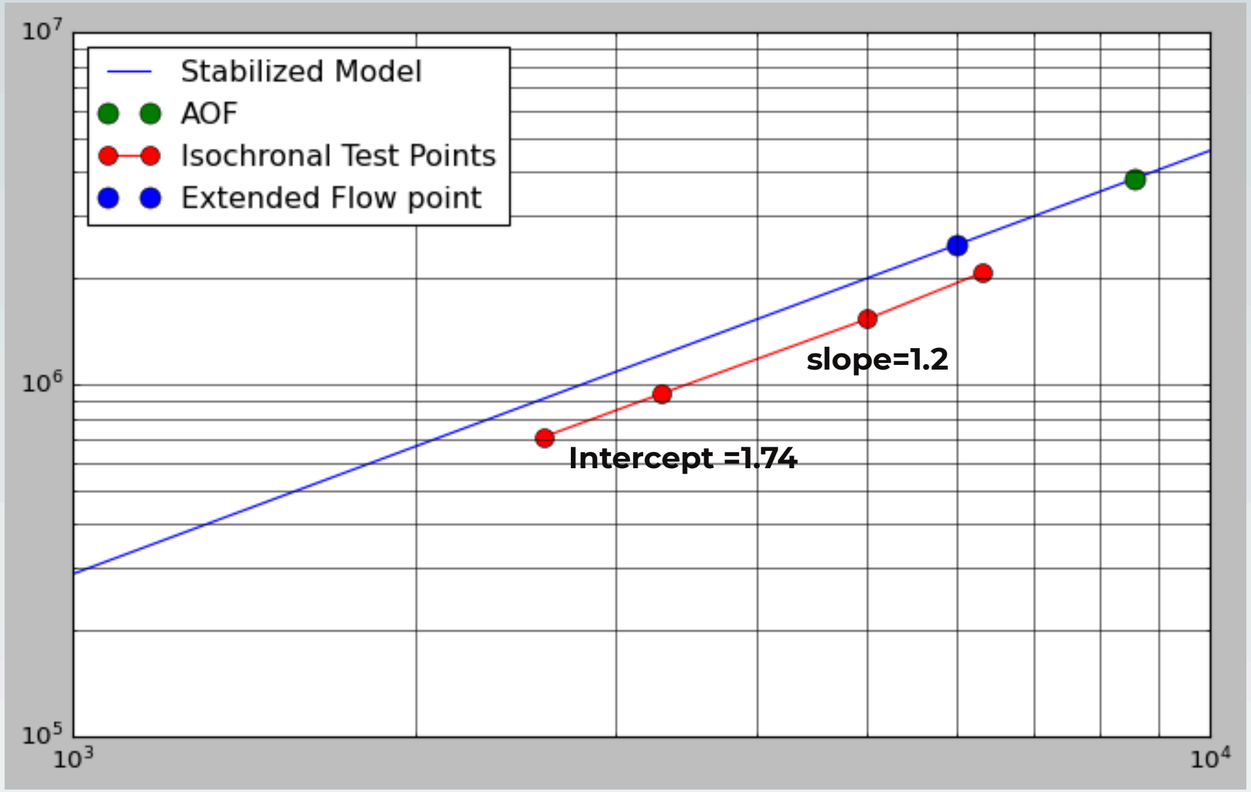
$$\log(\bar{p}^2 - p_{wf}^2) = \frac{1}{n}\log(Q_g) - \frac{\log(C)}{n}$$

n=1/slope
n=0.832
c=0.0287

$$C = \frac{Q_{g,st}}{(\bar{p}^2 - p_{wf}^2)_{st}^n}$$

Deliverability equation

$$Q_g = .029(\bar{p}^2 - p_{wf}^2)^{0.832}$$



Calculating Absolute open Flow by putting Pwf=14.7 psia

Absolute open flow is : 8574.717 MSCFD

Reference

- Class notes of well testing of Prof. Ashutosh Kumar sir(IIT ISM Dhanbad)
- Class notes of Well Performance of Prof. Rajeev Upadhyay (IIT ISM Dhanbad)
- Textbook Tarek Ahmed Advance Reservoir Engineering